

AMOGH GUPTA

Chapel Hill, NC

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Education

University of North Carolina at Chapel Hill

Aug 2025 – May 2027 (Expected)

Master of Science in Computer Science

Chapel Hill, NC

- Graduate student focused on NLP, AI, and research-driven development.
Teaching Assistant for Data 520: Research Methods for Socially Responsible AI
- Researcher in SAIL Lab, working with [Dr. Neil Gaikwad](#) on Responsible AI.
- Researching with [Prof. Snigdha Chaturvedi](#) on LLM Alignment.

Indian Institute of Information Technology Pune

Aug 2021 – June 2025

Bachelor of Technology in Computer Science Engineering

CGPA: 9.22

- Consistently ranked among the top 3 out of 175 students, reflecting academic excellence and commitment.
Sole student to score 10/10 SGPA in the 7th semester.

Indian Institute of Information Technology Pune

Aug 2023 – June 2025

Honors in Machine Learning (1st in Batch)

CGPA: 9.75

Work Experience

Amazon

May 2026 – Aug 2026

Software Development Engineer Intern: Alexa Audio

Seattle, WA

- * Built A3LS, an accent-audio testing service generating TTS across 13 accents and scoring per-accent ASR degradation (WER/AIER), via a Step Functions pipeline (Java Lambda, Coral/API Gateway) that fans out over tens of thousands of (utterance x accent) items per run.
- * Shipped a React/Cloudscape analytics dashboard over a CloudWatch metrics API, surfacing per-accent WER/AIER, latency percentiles, and GPU/concurrency, with skeleton loading and caching for instant loads.
- * Authored HLD/LLD design docs and drove cross-team design reviews and alignment to integrate an accent dimension for injecting synthetic audio into production ASR test runs.
- * Delivered across 80+ code reviews spanning 4 Brazil packages (Smithy, Java, CDK, React), owning CI/CD via Pipelines/Apollo.

Nomura

Jan 2025 – June 2025

Software Engineering Intern :React, Java Spring Boot, SQL, Jenkins, Agile

Mumbai, India

- * Worked on an internal budgeting and financial planning application used across multiple teams.
- * Resolved over 40 JIRA tickets in 6 months, focusing on performance improvements, bug fixes, and user experience.
- * Delivered impactful features and enhancements using React for frontend and Spring Boot for backend services.
- * Collaborated within an Agile team, using Jenkins for CI/CD and SQL for data-layer operations.

AIAssistant Startup

Aug 2024 – Dec 2024

Backend Intern: FastAPI, Python, Postman, AI

California, Remote

- * A dynamic startup focused on voice and agentic AI solutions.
- * Served as a backend intern developing solutions for multiple clients using Twilio, Deepgram, and ElevenLabs.
- * Developed prototypes for client demos, including call summarization, healthcare assistants, and lead-generation voice systems.

Ernst and Young

May 2024 – Aug 2024

Data and GenAI Intern

Pune, India

- * Worked on business and technical solutions for Fortune 100 clients driven by Generative AI. Gaining experience in cross-functional teams and technology consulting.
- * Developed a chatbot by implementing a **RAG Pipeline for over 10,000 documents** and improved search efficiency by 200% by creating metadata filters to narrow the search from 10,000 documents to 50 documents.

IIT Kharagpur

Apr 2024 – Aug 2024

Research Intern: Prompting and LLMs

Pune, India

- * Actively participated in red-teaming to identify and analyze safety limitations and vulnerabilities of LLMs.
- * Implemented several chains of thought to develop and improve multiple strategies for enhancing the strength of LLM models and their ethical applications. Made custom Python scripts to evaluate outputs in an automated manner.
- * Contributed towards generating new ideas for research, advancing the field, and supporting the project's objectives.

- * Successful in jail-breaking ChatGPT, Gemini, Nous-Hermes, Mixtral Instruct, Cohere Coral, and many more LLMs using custom-made multilingual prompts and cross-evaluated the outcomes by multiple sentimental models from HuggingFace.

IIT Indore

Jan 2024 – Apr 2024

Research Intern: Natural Language Processing

Pune, India

- * Contributed to an NLP project extracting critical information from disaster-related tweets by implementing advanced text summarization techniques, reducing a dataset of 3000 tweets to the crucial 40.
- * Large language models and integer linear programming methods were used to increase the accuracy and efficiency of the summarization process.
- * Engineered and participated in a flood detection project where we applied segmentation techniques to improve the classification outcomes and help identify and analyze the respective data for flooding.

RCI-Defence Research and Development Organisation (DRDO)

July 2023 – Aug 2023

Machine Learning Intern

Hyderabad, India

- * Explored and used advanced tools like Splunk and Kafka to improve the system's security measures.
- * Performed extensive log analysis and anomaly detection on over one million logs and located major outliers, which were then successfully reduced to a few hundred. This Proof of concept was further showcased to senior scientists to explore the possibilities for deployment in production.
- * Applied the One-Class SVM, KMeans, and Isolation Forests ML algorithms to improve detection accuracy.
- * Developed a dynamic web application powered by HTML, CSS, JavaScript, Bootstrap, and Flask.

Jio Platforms Limited

May 2023 – July 2023

Artificial Intelligence Intern

Mumbai, India

- * Worked on a project, 'Improving Operations with AI,' focusing on improving business operations using advanced AI.
- * Participated in designing and deploying recommendation algorithms, including Apriori, to enrich decision-making.
- * Designed a web application that gives a very engaging overview of the implemented features, presented later to the key stakeholders, showing the possible impact of the conducted project on operational efficiency.

Current Research

SHARD: Safe and Helpful Alignment via Self-Reframing Distillation [EMNLP 2026 Findings]

- * Large language models often struggle with sensitive prompts. They may refuse outright, provide generic safety boilerplate, or fail to address the user's legitimate informational needs that can be answered safely. We introduce SHARD, a self-reframing distillation method to improve safe-helpfulness.

Compounding Disadvantage: Auditing Intersectional Bias in LLM-Generated Explanations Across Indian and American STEM Education [ACM FAccT 2026]

- Accepted at the ACM Conference on Fairness, Accountability, and Transparency (FAccT) 2026. Auditing intersectional bias in AI-generated explanations across Indian and American STEM educational contexts, showing how disadvantages compound across language, caste, and social context rather than simply adding up.

Projects

ScholarSync, IIIT Pune | NodeJS, Websockets

Dec 2023 – Jan 2024

- Developed a centralized research collaboration portal using the MERN stack, Tailwind CSS, and Socket.IO, enabling cross-institutional collaboration among students and professors. The platform allows professors to post research opportunities and includes features like real-time chat, notifications for application updates, and dynamic profile management. Integrated a content-based recommendation system to suggest research papers based on user-entered keywords. All data is dynamically fetched through APIs for seamless user interaction.

CareerVaani, IIIT Pune | React, Bootstrap, NLP, Supervised Learning, API, MongoDB

Aug 2023 – Aug 2023

- Created an AI-based career counseling application for high school students, leveraging HTML, CSS, JavaScript, Bootstrap for the front end, and NLP, and supervised learning for career predictions. The application uses sentiment analysis and supervised algorithms to analyze questionnaire responses and provide tailored career recommendations. MongoDB was used for data storage, and API integration facilitated seamless interactions between the front and back end.

Technical Skills

GenAI & LLMs: LLM Alignment, Post-Training, Supervised Fine-Tuning (SFT), RLHF, DPO, LoRA/QLoRA, Inference Optimization, vLLM, DeepSpeed, Distributed Training, GPU/TPU Scaling, Prompt Engineering, RAG, Red-Teaming, Model Evaluation, Interpretability, Responsible AI

Programming & Frameworks: Python, C/C++, Java, PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LlamaIndex

AI/ML Expertise: Deep Learning, NLP, Large Language Models, Generative AI, Computer Vision (CNNs, Vision Transformers), Reinforcement Learning

Data & Databases: SQL, MongoDB, Vector Databases (FAISS, Pinecone), Pandas, NumPy, PySpark

Web & Dev Tools: Flask, FastAPI, Node.js, Express.js, Spring Boot, React, JavaScript, TypeScript, CSS, Bootstrap, REST APIs

Version Control & Deployment: Git/GitHub, Docker, Kubernetes, CI/CD (Jenkins), Cloud (AWS, GCP, Azure)